Drawing Straight Line Graphs

Exam code: 0607

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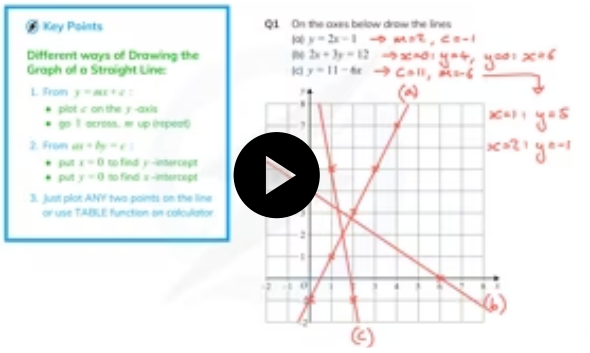

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Drawing linear graphs

How do I draw a straight line from a table of values?

Help me understand

- You may be given a **table of values** with **no** equation
- Use the x and y values to form a point with **coordinates** (x, y)
  - Then **plot** these points
  - Use a ruler to draw a **straight line** through them
    - All points should lie on the line
- For example
  - The points below are (-3, 0), (-2, 2), ... etc

x	-3	-2	-1	0	1	2	3
y	0	2	4	6	8	10	12

How do I draw a straight line using y = mx + c?

- Use the **equation** to create your own **table of values**
  - Choose points that are **spread out** across the axes given
- For example, plot $y = 2x + 1$ on axes from $x = 0$ to $x = 10$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

x	0	5	10
y	1	11	21

How do I draw a straight line without using a table of values?

- **Start** at the ***y*-intercept**, *c*
- Then, for every **1 unit** to the **right**, go up ***m* units**
  - *m* is the gradient
  - If *m* is **negative**, go **down**
  - If *m* is a **fraction**, remember that gradient is change in *y* divided by change in *x*
    - A gradient of $\frac{a}{b}$ would be a units up for every b units right
- This creates a sequence of **points** which you can then join up
  - Be **careful** of counting squares if axes have **different scales**
    - 1 unit might not be 1 square

Examiner Tips and Tricks

Always plot at least 3 points (just in case one of your end points is wrong!).

Worked Example

On the same set of axes, draw the graphs of $y = 3x - 1$ and $y = -\frac{3}{5}x + 3$.

Answer:

For $y = 3x - 1$, create a table of values

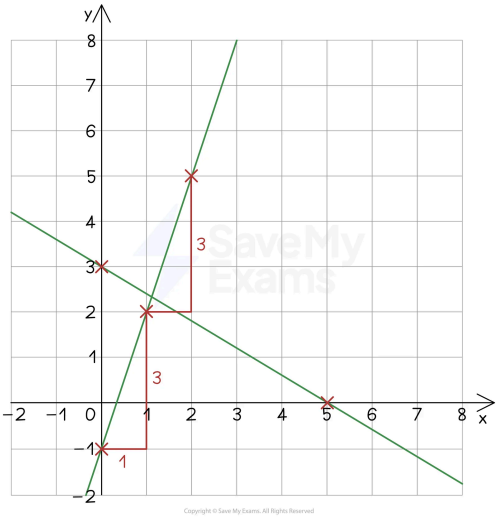
x	0	1	2
y	-1	2	5

Plot the points (0, -1), (1, 2) and (2, 5)
Connect with a straight line
Alternatively, start at the *y*-intercept (0, -1) and mark the next points 3 units up for every 1 unit to the right

For $y = -\frac{3}{5}x + 3$, create a table of values

Because of the fraction, $x = 5$ is a good point to include

Plot the points (0, -1), (1, 2) and (2, 5)
Connect with a straight line
Alternatively, start at the y-intercept (0, 3) and mark the next points 3 units down for every 5 units to the right



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